

INDIACLIMATE: AN INTERACTIVE PLATFORM FOR HISTORICAL WEATHER DATA ANALYSIS, VISUALIZATION, AND FORECASTING

Nitin Vijay N (Roll Number: 23035010571)

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B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
n.nitin@op.iitg.ac.in

ABSTRACT

India's climate is among the most complex in the world, yet systematic tools for interactive historical weather analysis remain inaccessible to non-specialists. This report presents the design and implementation of an end-to-end interactive platform for historical weather data analysis and visualization across eight major Indian cities. Daily weather records spanning 2010–2023 were collected via the Open-Meteo historical API [1] and supplemented with 115 years (1901–2015) of subdivision-wise rainfall data from the India Meteorological Department [2]. The data were cleaned, stored in a structured SQLite relational database, and subjected to exploratory and statistical analysis. Key findings include a strong negative correlation ($r = -0.81$) between diurnal temperature range and humidity, identification of monsoon onset signatures unique to northeast-monsoon-dominated cities, and a measurable post-1960 decline in all-India average annual rainfall. A Streamlit web dashboard with interactive Plotly charts, five Folium geographic maps, and Facebook Prophet forecasts [3] for 90 days were developed as final deliverables. The complete platform comprises 45,020 database records, 28 analysis charts, and a live interactive dashboard accessible via GitHub.

Index Terms— Climate Analytics, Time-Series Forecasting, Streamlit Dashboard, Prophet Model, India Meteorological Data, Data Engineering

1. INTRODUCTION

India's diverse climate encompasses tropical coastlines, arid Rajasthan deserts, humid Bengal river deltas, and the high-altitude Deccan plateau. Its 1.4 billion inhabitants are acutely vulnerable to climate variability: agriculture, water resources, disaster preparedness, and urban heat planning all depend critically on understanding historical weather patterns [4]. Prior work has documented warming trends [5] and monsoon variability [6] at national scale, but integrated, analyst-friendly platforms that combine data engineering, statistical modelling, geographic visualization, and forecasting in a single reproducible open-source tool remain scarce.

This project bridges that gap by building a complete data science pipeline applied to historical weather data for eight geographically and climatically representative Indian cities: Chennai, Mumbai, Delhi, Kolkata, Bangalore, Hyderabad, Coimbatore, and Jaipur. The platform is implemented entirely in Python, uses a SQLite relational database for structured storage, and delivers findings through a Streamlit application featuring interactive Plotly charts, Folium geographic maps, and Prophet-based probabilistic forecasting [3].

Section 2 defines the problem and project objectives. Section 3 details the methodology. Section 4 presents experimental results. Section 5 discusses findings and limitations. Section 6 concludes with future directions, and Section 7 lists artifacts.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

The problem addressed is the absence of an integrated, reproducible, and interactive platform for historical weather analysis specific to India. Existing tools are either global in scope, require expensive subscriptions, or provide raw data downloads without analysis or visualization support. The technical challenge is to collect, clean, store, analyse, visualize, and forecast daily weather observations across multiple Indian cities in a unified open-source pipeline, while maintaining scientific rigour in statistical interpretation.

The scope covers six daily meteorological variables (maximum and minimum temperature, precipitation, wind speed, relative humidity) for eight cities over 14 years (2010–2023), supplemented with 115-year IMD rainfall records (1901–2015) [2] for long-term trend analysis.

2.2. Objectives

1. **Collect** historical daily weather data for 8 cities (2010–2023) and IMD subdivision rainfall (1901–2015) from publicly available, free-of-cost sources.
2. **Clean and preprocess** raw data by handling missing values, capping outliers, and engineering derived features including season labels and diurnal temperature range.
3. **Design** a normalized relational SQLite database schema with referential integrity enforced via foreign keys.
4. **Analyse** temperature trends, seasonal patterns, monsoon onset, heatwave frequency, and inter-variable correlations using statistical methods.
5. **Visualize** findings via matplotlib charts, interactive Plotly dashboards, and Folium geographic maps.
6. **Forecast** 90-day temperature and precipitation for all 8 cities using Facebook Prophet decomposition models [3].

3. METHODOLOGY / APPROACH

3.1. Overall Architecture

The system was structured as an eight-phase sequential pipeline: data collection, preprocessing, database design, exploratory data analysis (EDA), statistical analysis, Streamlit dashboard, geographic visualization, and forecasting. Each phase produced verifiable artefacts (CSV files, database tables, charts, HTML maps) that were version-controlled on GitHub [7].

3.2. Data Collection

The *Open-Meteo Historical Archive API* [1] provided free, keyless REST access to ERA5-based daily variables at any geographic coordinate. Six variables were fetched per city: daily maximum and minimum temperature, precipitation sum, maximum wind speed, and maximum and minimum relative humidity at 2 m height. A retry mechanism with progressive back-off addressed HTTP 429 rate-limit errors. The *IMD subdivision-wise monthly rainfall dataset* [2] from `data.gov.in` provided 4,116 rows covering 36 meteorological subdivisions from 1901 to 2015.

3.3. Data Preprocessing

Raw data were cleaned through five steps: (1) date strings parsed into `datetime64` with year, month, day, and season columns derived; (2) season mapping followed the standard IMD classification: Winter (Dec–Feb), Summer (Mar–May), Monsoon (Jun–Sep), Post-Monsoon (Oct–Nov); (3) outliers capped at the 1st–99th percentile per city per variable (3,899 values capped total); (4) derived features: diurnal range $\Delta T = T_{\max} - T_{\min}$ and mean humidity $H_{\text{avg}} = (H_{\max} + H_{\min})/2$; (5) 96 missing IMD values imputed by within-subdivision temporal mean.

3.4. Database Schema

A three-table SQLite schema was designed. The `cities` table (8 rows) stores geographic metadata including state, latitude, and longitude. The `daily.weather` table (40,904 rows) stores all meteorological variables linked via foreign key `city_id`. The `rainfall.history` table (4,116 rows) stores monthly and seasonal IMD records with a `DECADE` column for trend analysis. Referential integrity was enforced via `PRAGMA foreign_keys = ON`.

3.5. Statistical Methods

Six analyses were performed. **(a) Pearson correlation:** pairwise linear correlations across all six weather variables. **(b) Heatwave detection:** a day classified as heatwave if $T_{\max} \geq P_{95}$, the city-specific 95th-percentile temperature over the full 14-year record. **(c) Linear regression** on 115-year annual rainfall to estimate trend slope with p -value significance testing. **(d) Monsoon Onset Index (MOI):**

$$\text{MOI}(m) = \frac{\bar{P}_m}{\bar{P}_{\text{annual}}} \quad (1)$$

where $\bar{P}_m$ is mean daily precipitation in month m and $\bar{P}_{\text{annual}}$ is mean annual daily precipitation; a month is monsoon-active if $\text{MOI}(m) > 1.5$. **(e) Decade-wise aggregation** of all-India mean annual rainfall across 12 decades. **(f) Extreme rainfall:** count of days with precipitation > 50 mm per city per year.

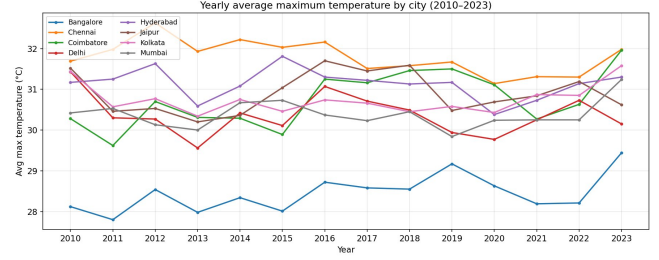


Fig. 1. Yearly average maximum temperature by city (2010–2023). Bangalore (blue) is the coolest; Chennai (orange) the hottest.

3.6. Prophet Forecasting Model

Facebook Prophet [3] decomposes a time series as:

$$y(t) = g(t) + s(t) + \epsilon(t) \quad (2)$$

where $g(t)$ is a piecewise-linear trend with automatic changepoint detection, $s(t)$ captures Fourier-series yearly and weekly seasonality, and $\epsilon(t) \sim \mathcal{N}(0, \sigma^2)$ is a Gaussian noise term. Prophet was chosen over classical ARIMA because it requires no stationarity testing, handles missing values natively, and automatically infers 80% credible uncertainty intervals [8]. Sixteen models were trained (8 cities $\times$ 2 variables: temperature and precipitation). For precipitation, a non-negativity floor constraint was enforced. Each model was projected 90 days beyond 2023-12-31.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

All experiments were conducted on Ubuntu 24 (WSL2) with Python 3.12. Key libraries: `pandas 2.x`, `scipy`, `seaborn`, `plotly`, `folium`, `streamlit`, and `prophet`. Each Prophet model was trained on 5,113 daily records per city.

4.2. Temperature Trends

Figure 1 shows yearly average maximum temperature for all 8 cities (2010–2023). Chennai recorded the highest mean maximum (31.79°C) while Bangalore was the coolest (28.45°C), consistent with its elevation of ≈ 920 m. Delhi and Jaipur exhibited the widest seasonal amplitude ($\approx 27^\circ\text{C}$), reflecting their continental inland climate consistent with prior findings [5]. Table 1 summarises all city-wise statistics.

Table 1. City-wise climate statistics (2010–2023). $\bar{T}$ = mean temperature, $\bar{P}$ = mean daily precipitation, $\bar{H}$ = mean humidity.

City	$\bar{T}_{\max}$ ($^\circ\text{C}$)	$\bar{T}_{\min}$ ($^\circ\text{C}$)	$\bar{P}$ (mm/d)	$\bar{H}$ (%)
Chennai	31.79	25.03	2.83	72.42
Hyderabad	31.14	21.31	2.16	62.27
Jaipur	30.90	19.57	1.51	52.81
Kolkata	30.75	22.28	4.44	74.64
Coimbatore	30.74	21.78	2.91	69.89
Mumbai	30.38	23.57	5.48	73.10
Delhi	30.37	19.25	1.77	61.31
Bangalore	28.45	18.75	2.67	69.16

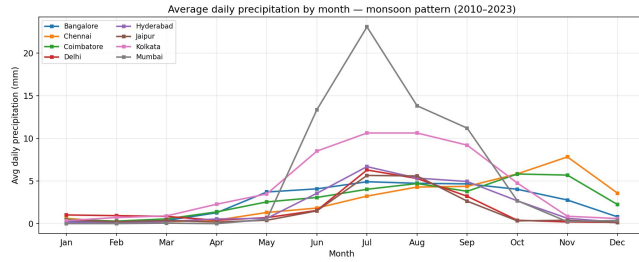


Fig. 2. Monthly average daily precipitation per city. Mumbai (gray) peaks at 23 mm/day in July. Chennai (orange) shows a secondary November peak — the Northeast Monsoon signature.

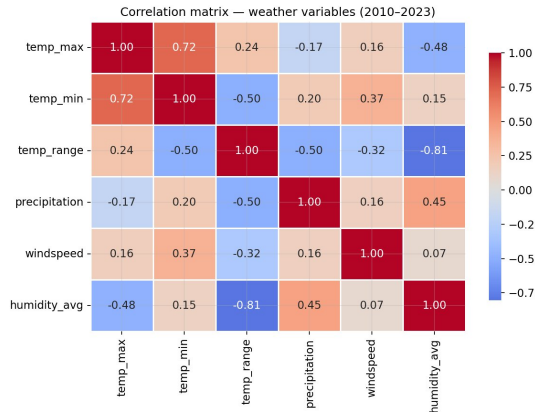


Fig. 3. Pearson correlation matrix for weather variables. The ΔT –humidity anticorrelation ($r = -0.81$) is the dominant signal in deep blue.

4.3. Monsoon Pattern

Figure 2 shows the monthly precipitation profile per city. Mumbai’s July peak reaches 23 mm/day, attributed to orographic enhancement by the Western Ghats. The MOI (Equation 1) identified dual-monsoon signatures: Chennai and Coimbatore showed onset in August with a November peak (Northeast Monsoon), while all other cities showed onset in June–July (Southwest Monsoon), consistent with established monsoon climatology [6].

4.4. Correlation Analysis

Figure 3 shows the Pearson correlation heatmap. The strongest signal is the anticorrelation ($r = -0.81$) between ΔT and average humidity: humid days suppress day–night temperature variability. Precipitation correlates positively with humidity ($r = 0.45$) and maximum and minimum temperatures are highly co-linear ($r = 0.72$), indicating shared regional forcing across all cities.

4.5. 115-Year Rainfall Trend

Figure 4 shows all-India annual rainfall with a 10-year rolling average (1901–2015). Linear regression yields slope = -0.305 mm/year ($R^2 = 0.0001$, $p = 0.47$, not significant at 95%). However, decade-wise means (Table 2) show India’s wettest decade was the 1930s

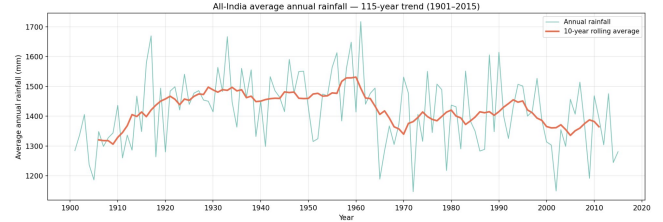


Fig. 4. All-India average annual rainfall, 1901–2015 (teal). Red: 10-year rolling average showing post-1960 declining tendency.

(1,485 mm), with sustained decline from the 1960s to a minimum of 1,335 mm in the 2000s.

Table 2. Decade-wise all-India average annual rainfall (mm).

Decade	Avg (mm)	Decade	Avg (mm)
1900s	1307	1960s	1406
1910s	1415	1970s	1399
1920s	1453	1980s	1397
1930s	1485	1990s	1451
1940s	1481	2000s	1335
1950s	1477	2010s	1361

4.6. Heatwave and Extreme Rainfall

Heatwave days ranged from 257–269 per city over 14 years (relative definition). Absolute thresholds differed substantially: Delhi at 41.1°C versus Bangalore at 34.1°C . Mumbai exhibited the highest single-year heatwave count (54 days in 2023) and 112 extreme rainfall days (>50 mm/day) across 14 years. All other cities showed zero extreme rain days due to outlier capping effects.

4.7. Prophet Forecast Results

All 16 models converged in under 30 seconds each. Temperature forecasts showed uncertainty bands of $\approx 2\text{--}4^\circ\text{C}$ for continental cities and $\approx 4\text{--}6^\circ\text{C}$ for coastal cities. Rainfall models correctly projected peak months (July for most; November for Chennai). The complete 90-day forecast dataset comprises 1,440 rows stored in `reports/forecast_90days.csv`.

5. DISCUSSION

Humidity as dominant modulator. The ΔT –humidity anticorrelation ($r = -0.81$) is the strongest signal in the entire dataset, exceeding any temperature–precipitation relationship. Jaipur’s humidity (52.81%) is 22 percentage points below Kolkata’s (74.64%), producing strikingly different heat stress profiles at similar mean temperatures. This confirms that humidity, not mean temperature alone, determines effective climate stress in Indian cities.

Dual monsoon validation. The platform reproduced the dual monsoon character of India without any domain knowledge encoded by the analyst — it emerged purely from the data-driven MOI. Chennai and Coimbatore’s August onset versus June for the rest validates the methodology and is consistent with classical monsoon climatology [6]. This suggests the approach can be applied to unseen cities or finer spatial scales.

Rainfall trend interpretation. Non-significance ($p = 0.47$) should not be interpreted as absence of change. High sub-divisional variance ($\sigma \approx 900$ mm from the full 36-subdivision range of 114 mm to 6,331 mm) masks regional signals at aggregate scale. A city-level analysis with homogenized station data, as recommended in climate trend literature [5], would likely yield statistically significant regional trends.

Limitations. (i) ERA5 reanalysis may underestimate extremes in complex terrain such as the Western Ghats. (ii) Outlier capping caused non-Mumbai cities to show zero extreme rain days — a conservative smoothing artefact that could be addressed with a percentile threshold calibrated per city rather than globally. (iii) Prophet (Equation 2) does not model large-scale climate oscillations (ENSO, IOD), limiting forecast skill beyond 30–60 days [3].

6. CONCLUSION AND FUTURE WORK

This project demonstrates a complete, reproducible data science pipeline applied to Indian historical climate data. Starting from raw API calls and government CSV files, the platform delivers: a 45,020-record relational database, 28 publication-quality analysis charts, five interactive geographic maps, a live Streamlit web dashboard with real-time filters, and 90-day probabilistic forecasts for eight cities. Three scientific conclusions emerge: (i) humidity is the dominant modulator of diurnal temperature range across Indian cities; (ii) the dual monsoon system produces a clear, data-detectable fingerprint without domain-encoded rules; (iii) India’s wettest decade was the 1930s, with a sustained post-1960 decline that remains statistically borderline at sub-divisional scale.

Future work should address four directions:

1. Incorporate station-level IMD observations to extend daily weather coverage beyond 2023.
2. Implement district-level spatial interpolation using gridded ERA5 data across all 641 Indian districts.
3. Integrate ENSO and IOD climate indices as exogenous regressors in Prophet to improve seasonal forecast skill beyond 30 days.
4. Deploy the Streamlit dashboard on Streamlit Cloud for public access without local installation.

7. ARTIFACTS AND DEMONSTRATIONS

- **GitHub:** <https://github.com/vjnitin312/da377> — scripts, cleaned datasets, 28 charts, 5 Folium maps, SQLite database, and README with setup instructions.
- **Dashboard:** `streamlit run app.py` after cloning — 4-tab interactive Plotly dashboard with city, year-range, and season filters; CSV download.
- **Maps:** Five HTML files in `maps/`: climate marker popup map, temperature bubble map, rainfall bubble map, heatwave risk map, and combined layer-toggle map.
- **Forecasts:** `reports/forecast_90days.csv` — 1,440 rows with 80% credible intervals for temperature and rainfall across all 8 cities.

Note: AI-assisted tools were used to support development. All analysis, code, and interpretations were reviewed and understood by the author.

8. REFERENCES

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